

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VIII
PAPER –1
TEST DATE: 14-08-2022

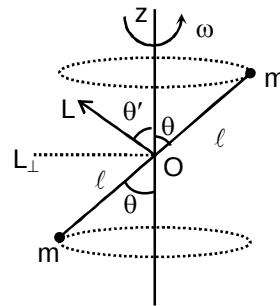
ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B
Sol. Mass defect
 $\Delta m = -0.00129 \text{ u}$
Energy = 0.00129×931
= 1.20 MeV
2. C
Sol. Buoyant force is real force and will be same in all reference frames.
3. D
Sol. Pressure variation is maximum at ends, so there will be pressure antinodes at ends.
4. C
Sol. One component of velocity is along +ve y-axis
So, path will be helical. Path of particle will touch y-axis after every $\frac{2\pi m}{qB}$.
5. A, B, C
Sol. Angular momentum of system about O is L
 $|L| = 2m\omega\ell^2\cos\theta'$
 $L_z = L \cos\theta' = 2m\omega\ell^2\cos^2\theta'$
 $L_z = 2m\omega\ell^2\sin^2\theta = mv\ell\cos\theta'$
6. A, B
Sol. L.C. = 1 msd = 1 vsd



$$0.3 = 1 - \frac{B}{A}$$

$$\frac{B}{A} = 1 - 0.3$$

$$\frac{B}{A} = 0.7 \quad \dots(i)$$

$$A = 15$$

$$B = 15 \times 0.7 = 10.5, \text{ also for } A = 10, B = 7 \text{ mm}$$

7. B, C

Sol. Temperature of gas at C is minimum as temperature of gas from B $\rightarrow$ C continuously decreases.

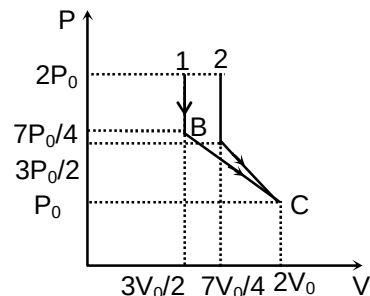
Heat dQ, in process,

$$dQ_1 = \frac{V_0}{2} \left(\frac{P_0 + \frac{7P_0}{4}}{2} \right) + n \frac{3}{2} R \left[\frac{2P_0 V_0}{nR} - \frac{3P_0 V_0}{nR} \right] = \frac{-13P_0 V_0}{16}$$

In process 2

$$dQ_2 = \frac{V_0}{4} \left[\frac{P_0 + \frac{3P_0}{2}}{2} \right] + n \times \frac{3}{2} R \left[\frac{2P_0 V_0}{nR} - \frac{21P_0 V_0}{8nR} \right]$$

$$= \frac{5P_0 V_0}{16} - \frac{15P_0 V_0}{16} = \frac{-5P_0 V_0}{8}$$



8. A, C, D

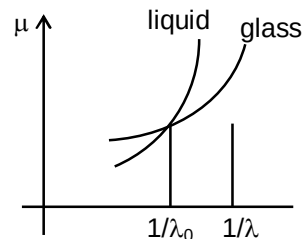
Sol. Acceleration a P will be zero, if current in solenoid i is constant

$$\text{Velocity of charge at point Q, } v = \sqrt{\frac{qR^2}{m} \frac{dB}{dt} \theta} = \sqrt{\frac{2 \times 10^{-6} \times (1)^2}{10^{-6}} \times \frac{\pi}{3} \times 5} = 3.2 \text{ m/s}$$

9. A, B, C, D

Sol. For smaller $\lambda < \lambda_0$, $\mu_\ell > \mu_g$

Light inside prism, and emergent light through AC will deviate away from normal at point of incidence.



10. A, B, C

$$\text{Sol. } \frac{1}{2}mv^2 = 3 \frac{1}{4\pi\epsilon_0} \times \frac{Q}{\frac{a}{\sqrt{3}}}$$

$$\Rightarrow v = \sqrt{\frac{3\sqrt{3}Q}{2\pi\epsilon_0 ma}}$$

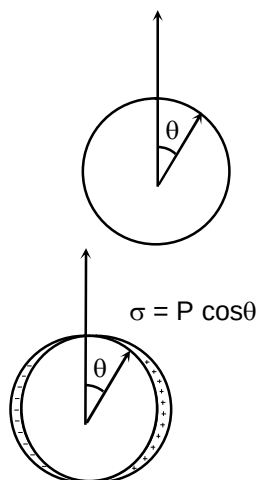
Section – B

11. 4

Sol. Surface charge density, $\sigma = \vec{P} \cdot \hat{n} = P \cos \theta$
 This charge distribution can be represented by
 $\sigma = P \cos \theta$
 Volume charge density, $\rho_b = 0$

$$\text{Electric field, } E = \frac{P}{3\epsilon_0} = \frac{4\pi P}{3 \times 4\pi\epsilon_0}$$

$$= \frac{4\pi \times \frac{10^{-9}}{\pi} \times 9 \times 10^9}{3} = 12 \text{ N/C}$$



12. 4

Sol. $\omega t - kx = \sin^{-1} \frac{y_1}{A}$

$$\omega t - ky = \sin^{-1} \frac{y_2}{A}$$

$$\Rightarrow k(x - y) = \sin^{-1} \left(\frac{y_2}{A} \right) - \sin^{-1} \left(\frac{y_1}{A} \right)$$

$$\sin^{-1} \left(\frac{y_2}{A} \right) = k(x - y) + \sin^{-1} \left(\frac{y_1}{A} \right)$$

$$\frac{y_2}{A} = \sin \left[k(x - y) + \sin^{-1} \left(\frac{y_1}{A} \right) \right]$$

$$\frac{y_2}{A} = \sin k(x - y) \sqrt{1 - \frac{y_1^2}{A^2}} + \cos[k(x - y)] \left(\frac{y_1}{A} \right)$$

$$\text{For } \cos k(x - y) = 0$$

$$k(x - y) = \pm \left(n' \pm \frac{1}{2} \right) \pi$$

$$(x - y) = \pm \left(n' \pm \frac{1}{2} \right) \frac{\lambda}{2}$$

$$y = x \pm \left(n' \pm \frac{1}{2} \right) \frac{\lambda}{2}$$

$$y = x \pm \left(\frac{n'}{2} \pm \frac{1}{4} \right) \lambda$$

13. 2

Sol. Disc will be in kinematic equilibrium. Its point of contact with ground will move with constant velocity, so net torque about it must be zero.

Tension in string must pass through point of contact C.

$$\angle BOC = (180 - 2\alpha)$$

$$\text{Also, } \theta + 2\alpha - 90 + 180 - \alpha = 180$$

$$\Rightarrow \theta + \alpha = 90^\circ$$

$$\alpha = (90 - \theta) \quad \dots(i)$$

$$\triangle OBC, \frac{\sin \alpha}{R} = \frac{\sin(180 - 2\alpha)}{y}$$

$$\frac{\sin \alpha}{R} = \frac{2 \sin \alpha \cos \alpha}{y}$$

$$y = 2R \cos \alpha = 2R \cos (90 - \theta) = 2R \sin \theta \quad \dots(ii)$$

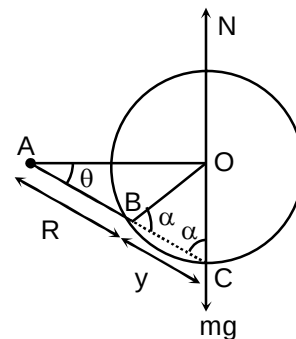
$\triangle AOC$:

$$\sin \theta = \frac{R}{R + 2R \sin \theta}$$

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

$$\sin \theta = \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4}$$

$$\sin \theta = \frac{1}{2}$$



Section - C

14. 0.02

Sol. $i = i_0 \cos \omega t \quad \dots(i)$

$$V_L = V_0 \cos \left(\omega t + \frac{\pi}{2} \right) = -V_0 \sin \omega t \quad \dots(ii)$$

$$V_C = V'_0 \cos \left(\omega t - \frac{\pi}{2} \right) = V'_0 \sin \omega t \quad \dots(iii)$$

$$\frac{V_L}{i} = -\frac{V_0}{i_0} \tan \omega t$$

$$\frac{1}{1} = \frac{-50 \times 20 \times 10^{-3} i_0}{i_0} \tan(50t_0)$$

$$\tan(50t_0) = -1 \Rightarrow t_0 = \frac{3\pi}{4 \times 50} \text{ sec} = \frac{3\pi}{200} \text{ sec}$$

When $V_{PQ} = 0$

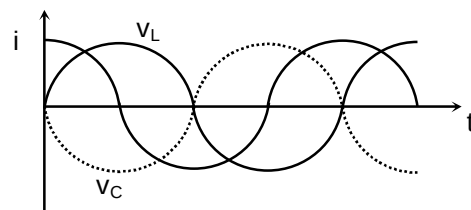
$$V_L + V_C = 0$$

$$\sin \omega t (V'_0 - V_0) = 0$$

$$\sin \omega t = 0$$

$$t = \frac{\pi}{50}$$

$$x = t - t_0 = \frac{\pi}{50} - \frac{3\pi}{200} = \frac{\pi}{200} \approx 0.02 \text{ sec}$$



15. 0.01

Sol. Voltage across L and C will be of opposite polarity

$$V_L = L \frac{\Delta i}{\Delta t} = 20 \times 10^{-3} \times 10 = 0.2 \text{ volt} = \text{constant}$$

Current in branch at time t

$$i = 1 - 10t$$

$$\frac{dq}{dt} = 1 - 10t$$

$$q = (t - 5t^2)$$

For net voltage be zero

$$0.01 = t - 5t^2$$

$$5t^2 - t + 0.01 = 0$$

$$\Rightarrow t = 0.01 \text{ sec}$$

16. 1.20

17. 2.08

Sol. (for Q.16-17)

$$\frac{1.2}{v_1} - \frac{1}{-40} = \frac{1.5-1}{20} + \frac{1.2-1.5}{-24}$$

$$\frac{1.2}{v_1} = \frac{1}{40} + \frac{3}{240} - \frac{1}{40}$$

$$v_1 = 80 \text{ cm}$$

For concave lens

$$\frac{1}{v_2} - \frac{1}{30} = \frac{1}{20}$$

$$\frac{1}{v_2} = \frac{1}{20} + \frac{1}{30}$$

$$v_2 = 12 \text{ cm}$$

$$\text{For } L_1, m = \frac{v}{u} = \frac{12}{30}$$

$$\frac{h_i}{h_o} = \frac{2}{5}$$

$$h_i = \frac{2}{5} \times 1 = 0.4 \text{ mm}$$

Distance between image formed by L_1 and $L_2 = 1.2 \text{ mm}$

$$\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 0.5}{1.2 \times 10^{-3}} = \frac{25}{1.2} \times 10^{-4} \text{ m}$$

18. 21.00

19. 15.00

Sol. (for Q.18-19)

Charge on plates when k is open

Charge on inner face of plate B = $\frac{6}{2} = 3 \mu\text{C}$ and charge on outer surface of plate B = $3 \mu\text{C}$.

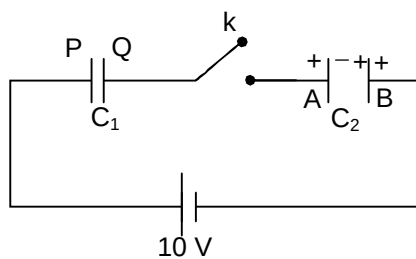


Figure-(1)

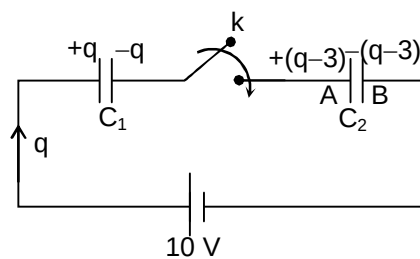


Figure-(2)

For figure (ii)

$$10 - \frac{q}{3} - \frac{q-3}{6} = 0$$

$$60 - 2q - q + 3 = 0$$

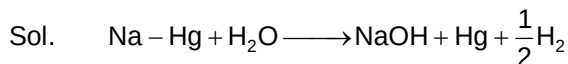
$$q = 21 \mu\text{C}$$

Charge on plate P = $21 \mu\text{C}$

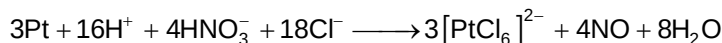
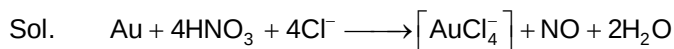
Charge on plate B = $[-(21 - 3) + 3] \mu\text{C} = -15 \mu\text{C}$

Chemistry**PART – II****Section – A**

20. D



21. C



22. A

Sol. Concentration of $[\text{H}^+]$ at anode = 10^{-8} M
 $\therefore$ concentration of $[\text{H}^+]$ at cathode = 10^{-7} M

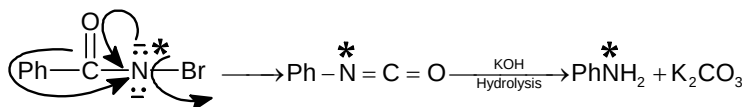
$$\text{because } \text{pH} = \frac{1}{2}[pkw + pka - pkb] = \frac{1}{2}[14 + 4.74 - 4.74] = 7.$$

$$\text{so } [\text{H}^+] = 10^{-7} \text{ M}$$

$$E = 0 - \frac{0.059}{1} \log \frac{10^{-8}}{10^{-7}} = E = 0.059 \text{ V}$$

23. B

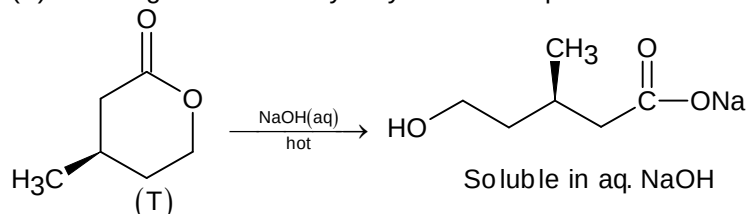
Sol.



Intramolecular rearrangement takes place.

24. A, C, D

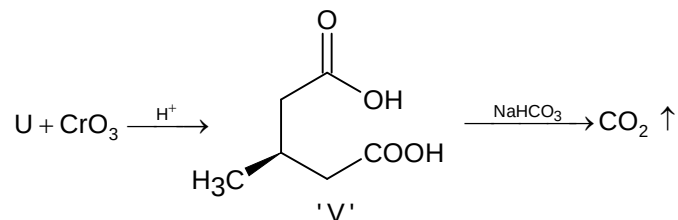
Sol. (A) T undergoes an ester hydrolysis in hot aqueous alkali as



(B) LiAlH_4 reduces ester to alcohol as 'U' no chiral carbo, optically inactive

(C) U on treatment with excess of acetic anhydride form a diester

(D) U on treatment with $\text{CrO}_3 / \text{H}^+$ undergo oxidation to diacid which gives effervescence with NaHCO_3



But $r_n + 1 - r_n = r_n - 1$

$$(n+1)^2 - n^2 = (n-1)^2$$

$$n = 4$$

32. 6

Sol. $\Delta E = \Delta H = \Delta T = q = P_{\text{ext}} = \Delta S_{\text{surr}} = 0$

$$\Delta S_{\text{sys}} = \Delta S_{\text{total}} = +ve$$

$$\Delta G_{\text{sys}} = -Ve$$

Section – C

33. 0.60

34. 1.40

Sol. **(for Q. 33 to 34)**

$$\frac{P_A^0}{2} + \frac{P_B^0}{2} = 1 \Rightarrow P_A^0 + P_B^0 = 2$$

$$\frac{P_A^0}{4} + \frac{3P_B^0}{4} > 1 \text{ atm} \quad P_A^0 + 3P_B^0 > 4 \text{ atm}$$

$$\text{and } \frac{P_A^0}{8} + \frac{3P_B^0}{8} + \frac{4P_C^0}{8} = 1 \Rightarrow P_A^0 + 3P_B^0 + 4P_C^0 = 8 \text{ atm.}$$

$$\Rightarrow P_A^0 + 3P_B^0 = (8 - 4 \times 0.8) \text{ atm} = 4.8 \text{ atm.}$$

35. 5.00

36. 4.00

Sol. **(for Q. 35 to 36)**

$$[Ma_2b_2c_2] - \text{Geometrical isomers} = 5$$

$$[M(AB)_3] - \text{Optical isomers} = 4$$

All are optical active isomers.

37. 4.50

38. 5.20

Sol. **(for Q. 37 to 38):**

$$\text{Distance between nearest neighbour} = \frac{\sqrt{3}a}{2}$$

$$\text{Distance between next nearest neighbour} = a.$$

Mathematics

PART – III

Section – A

39. A

$$\text{Sol. } \frac{2 \sin 40^\circ + \sin 20^\circ}{\cos 20^\circ \cdot \cos 30^\circ} = \frac{2 \sin(60^\circ - 20^\circ) + \sin 20^\circ}{\cos 20^\circ \cdot \cos 30^\circ}$$

40. C

$$\begin{aligned} \text{Sol. } \int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx &= \int \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx \\ &= \int \frac{\tan^2 x \cdot \sec^2 x}{(1 + \tan^3 x)^2} dx \text{ put } 1 + \tan^3 x = t \\ &= \frac{1}{3} \int \frac{dt}{t^2} \Rightarrow -\frac{1}{3t} + c = -\frac{1}{3(1 + \tan^3 x)} + c \end{aligned}$$

41. D

$$\begin{aligned} \text{Sol. } (5x-4) \begin{vmatrix} 1 & 2x & 2x \\ 1 & x-4 & 2x \\ 1 & 2x & x-4 \end{vmatrix} &= (5x-4) \begin{vmatrix} 1 & 2x & 2x \\ 0 & -(x+4) & 0 \\ 0 & 0 & -(x+4) \end{vmatrix} = (5x-4)(x+4)^2 = (A+Bx)(x-A)^2 \\ \Rightarrow A &= -4, B = 5 \end{aligned}$$

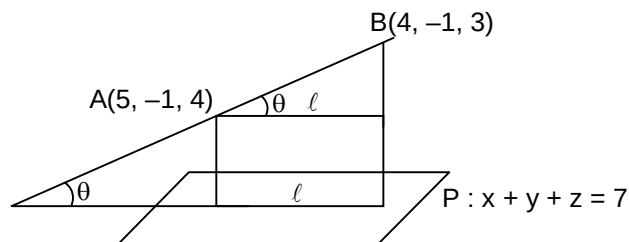
42. A

Sol. D.R's of AB = (1, 0, 1)
D.R's of normal to plane = (1, 1, 1)

$$\cos\left(\frac{\pi}{2} - \theta\right) = \frac{1+1}{\sqrt{2}\sqrt{3}} = \frac{\sqrt{2}}{\sqrt{3}} = \sin \theta$$

$$AB = \sqrt{2}$$

$$\text{Length of projection} = AB \cos \theta = \sqrt{\frac{2}{3}}$$



43. A, B, C

$$\text{Sol. } \frac{\pi}{2} - \tan^{-1}(na), a > 0, a = 0, a < 0$$

44. A, C

$$\text{Sol. } I.F = \frac{1}{\sin^3 x}, y = -2 \sin^2 x + 4 \sin^3 x, y\left(\frac{\pi}{2}\right) = 2$$

45. B, C, D

$$\text{Sol. } \text{Locus is } y^2 = 4a(x - 4a)$$

46. B, D

$$\text{Sol. } \text{Let } \sqrt{x} = \cos 2\theta$$

$$I = \int_0^{\frac{\pi}{4}} \cot \theta \cdot \sin 4\theta d\theta, J = \int_0^{\frac{\pi}{4}} \tan \theta \cdot \sin 4\theta d\theta$$

47. A, C, D
 Sol. $A = I - B$
 $A^2 = I^2 + B^2 - 2B = I - B = A$
 $AB = 0$
 $BA = 0$

48. A, C
 Sol. $0 < \sin x < 1, 0 < \cos x < 1$
 $\sin^n x + \cos^n x = 1 \quad n = 2$
 $> 1 \quad n < 2$
 $< 1 \quad n > 2$

Section – B

49. 5
 Sol. $\sum (z)^r = \sum_{r=1}^{50} \pi(z - z_r)$. Take log and differentiate

50. 2
 Sol. $\sqrt{\frac{2}{6}} = \frac{2}{\sqrt{3}}, b = \frac{3}{2}$

51. 3
 Sol. In ΔABC , $\sum \cot \frac{A}{2} = \left(\cot \frac{A}{2} \right) \left(\cot \frac{B}{2} \right) \left(\cot \frac{C}{2} \right)$
 $AM \geq GM; \cot^2 \frac{B}{2} \geq 3$

Section – C

52. 864.00
 53. 6912.00
 Sol. **(for Q. 52.-53)**
 All roots are negative. $AM \geq GM$

54. 0.00

55. 2.00
 Sol. **(for Q. 54.-55):**
 $f(x) = e^x \int_0^1 e^t f(t) dt = Ae^x; A = \int_0^1 e^t f(t) dt; f(t) = Ae^t, A = 0$
 $f(x) = 0, f(0) = 0$
 $g(x) = e^x \int_0^1 e^t g(t) dt + x = Be^x + x; B = \int_0^1 e^t g(t) dt; B = \frac{2}{3 - e^2}$
 $g(x) = \left(\frac{2}{3 - e^2} \right) e^x + x; g(0) = \frac{2}{3 - e^2} \Rightarrow \left[f(0) - \frac{1}{g(0)} \right] = \left[\frac{e^2 - 3}{2} \right] = 2$

56. 8.00

57. 20.00

Sol. **(for Q. 56.-57):**

Draw the figure: AT and BT are radical axis to C_3 and C_1 and C_3 and C_2

$$r_3 = TA \tan\left(\frac{\theta_1 + \theta_2}{2}\right) = 4 \left(\frac{\frac{1}{2} + \frac{3}{4}}{1 - \frac{1}{2} \cdot \frac{3}{4}} \right) = 8$$

$TA = TB = TD = 4$: T is radical centre

$$\tan\left(\frac{\theta_1}{2}\right) = \frac{1}{2} ; \tan\left(\frac{\theta_2}{2}\right) = \frac{3}{4}$$

TC_3 will be the diameter of circumcircle of $\triangle TAB$

$$TC_3 = \frac{TA}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)} = 4\sqrt{5} \therefore 20\pi$$